Graphical solutions to linear equations or inequalities

1 For each of the following equations:

i Solve for the value of x that satisfies the equation.

ii To verify the solution graphically, which two straight lines would need to be graphed?

iii Graph the two lines in part (b) on the same plane.

iv Hence find the value of x that satisfies the two linear equations simultaneously.

a $\frac{3x}{5} - 1 = 2$

b $2(x-1) - 3 = 7$

c $2x = 8 - 2x$

d $4x + 3 = 2x - 1$

e $2x + 3 = 6 - x$

f $2(2x+2) = -3(x+1)$

2 Using a graphing calculator, solve the following equations:

a $2x - 7 = -2x + 1$

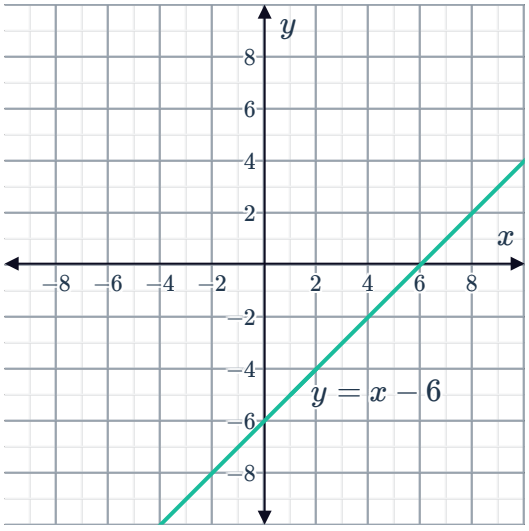
b $3x + 6 - x = 7x - 4$

c $8x - 3(7 - 2x) = 6(x - 4) + 5x + 9$

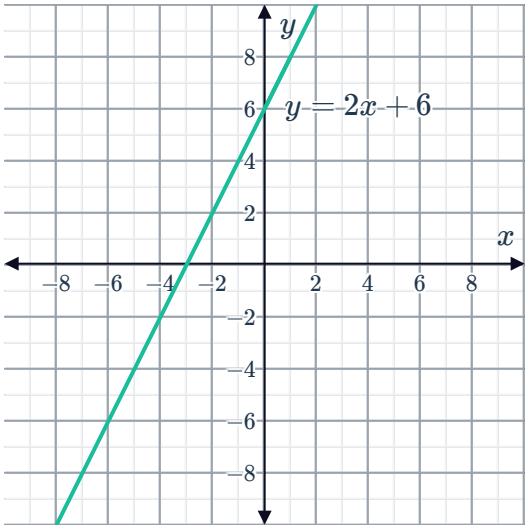
d $-4(3x + 8) = -7x + 8 - 5x$

3 Using the graphs, state the solution of the following inequalities:

a $x - 6 < 0$

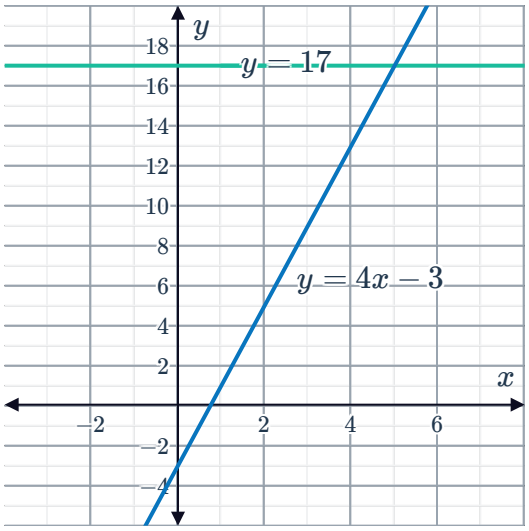


b $2x + 6 \geq 0$

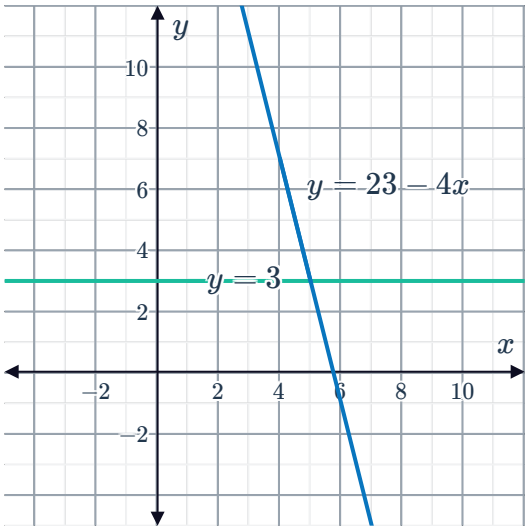


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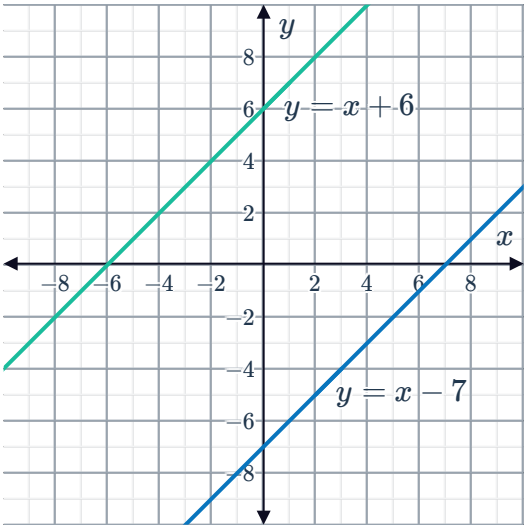
c $4x - 3 < 17$



d $23 - 4x < 3$



4 Using the graphs of $y = x + 6$ and $y = x - 7$, find how many solutions the inequality $x + 6 \geq x - 7$ has.



5 To solve the inequality $x \leq 2x - 3$, Christa graphed $y = x + 3$. Find the other line she needs to graph to be able to solve the inequality graphically.

6 To solve the inequality $x \leq \frac{x - 3}{4} - 1$, Tracy graphed $y = x - 3$. Find the other line she needs to graph to be able to solve the inequality graphically.

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7 State whether the following inequalities can be solved graphically using the graphs of $y = 4x + 3$ and $y = 5 - x$:

a $(4x + 3) - (5 - x) \leq 0$

b $(4x + 3) + (3 - x) < 0$

c $\dfrac{4x + 3}{5} + x > 0$

d $3x + 8 < 0$

e $5x > 2$

8 Consider the linear function: $f(x) = \dfrac{-3}{2}x - 4$

a Write a linear function $g(x)$ such that $f(x) = g(x)$ has exactly one solution.

b With the $g(x)$ determined in part (a), find the exact solution of $f(x) = g(x)$.

9 Consider the linear equation $2y = -3x + 8$:

a Graph the linear equation by determining any two points on the line.

b This linear equation contains the point $(2, k)$. Find the value of k.

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